

Sectoral and Market Responses to GST Policy Shocks in India: Evidence from VARX and QVAR Models”

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Chapter 1

Abstract

This study undertakes a comprehensive two-pronged analysis of the economic implications of adjustments in the goods and Services Tax (GST) rate in India. The research integrates both financial market and macroeconomic perspectives to provide a holistic understanding of the reform's impact.

The first section of this study examines the dynamic impact of Goods and Services Tax (GST) rate cut announcements and GST council meetings on sectoral stock indices in India. Using a Vector Autoregressive model with exogenous variables (VARX), the analysis captures both the direct and spillover effects of fiscal policy signals across major sectors. Impulse Response Functions (IRFs) and Forecast Error Variance Decomposition (FEVD) provide insights into the temporal magnitude and persistence of policy-induced shocks. The results highlight the heterogeneous sensitivity of sectoral returns to tax policy interventions. The second component adopts a macroeconomic lens, applying a Two-Stage Least Squares (2SLS) instrumental variable strategy to estimate the causal relationship between fiscal shocks—proxied by monthly GST revenue—and two key aggregates: private final consumption expenditure and gross domestic savings. This econometric design mitigates endogeneity concerns arising from the simultaneous interaction between tax revenues and household spending.

Empirical results reveal heterogeneous responses across sectors, with consumption-oriented industries displaying pronounced sensitivity to GST rate cut announcements. At the macroeconomic level, the findings indicate a statistically significant and economically meaningful positive effect of exogenous GST revenue shocks on private consumption, consistent with the presence of a strong fiscal multiplier in the Indian context.

By bridging high-frequency market dynamics with macroeconomic outcomes, this study contributes to the growing literature on fiscal policy transmission in emerging economies. It offers evidence-based insights for policymakers on the dual dimensions of tax reform—its

influence on financial market expectations and its role in shaping aggregate demand.

Chapter 2

Introduction and Background

2.1 Introduction

The implementation of the Goods and Services Tax (GST) in India on July 1, 2017, marked a major milestone in the country's fiscal history. By replacing a complex network of indirect taxes at both central and state levels with a unified tax system, GST aimed to simplify compliance, reduce tax cascading, and improve efficiency in revenue collection. Beyond its fiscal objectives, GST has significant implications for financial markets and sectoral economic activity, influencing inflation, consumption demand, and sectoral risk-return dynamics.

Financial markets often respond not only to actual policy changes but also to anticipatory signals regarding future fiscal adjustments. Investors and market participants closely monitor GST Council meetings and announcements, especially rate-cut decisions, as these events can affect expectations about inflation, sectoral demand, and market volatility. Understanding these responses is crucial for policymakers, as they provide insights into market efficiency, the transmission of fiscal shocks, and potential unintended consequences.

This study examines the short- and medium-term effects of GST-related policy actions on Indian equity markets and macroeconomic indicators. Using a combination of event study analysis, multivariate time-series modeling (VARX and QVAR), impulse response functions (IRFs), and forecast error variance decomposition (FEVD), it investigates how GST shocks propagate across sectors, interact with risk premia, and influence market dynamics.

2.2 Background

2.2.1 GST in the Indian Context

Before GST, India's indirect tax system was fragmented, with multiple levies such as VAT, excise duty, and service tax, resulting in inefficiencies, higher compliance costs, and opportunities for tax arbitrage. The GST reform introduced a dual system: Central GST (CGST) and State GST (SGST) for intra-state transactions, and Integrated GST (IGST) for inter-state trade. Harmonizing rates and procedures, GST improved transparency, simplified compliance, and expanded the tax base.

GST policy actions—particularly Council meetings especially on September 3, 2025 and rate reduction announcements—affect market expectations and economic demand. Rate changes or tax rationalizations directly influence the prices of goods and services, thereby shaping inflationary pressures and sectoral consumption patterns. Investors interpret these policy signals as indicative of broader macroeconomic trends, causing short-lived adjustments in market volatility and sectoral returns.

The economic effects of GST can be summarized as follows:

- **Inflation and Price Stability:** Changes in GST rates affect final consumer prices, influencing inflation and shaping aggregate demand responses.
- **Market Efficiency and Anticipatory Behavior:** Equity markets incorporate both actual and anticipated GST policy changes. Observing sectoral reactions to Council announcements provides insight into the speed and efficiency of information assimilation.
- **Policy Transmission through Rate Adjustments:** GST rate cuts or rationalizations can moderate demand-side pressures and stabilize inflation, highlighting the importance of clear and timely communication by the GST Council.

2.2.2 Market Responses to Fiscal Policy

Financial markets process policy information rapidly, and several theoretical channels explain how fiscal shocks affect asset prices:

- **Efficient Market Hypothesis (EMH):** Markets quickly incorporate publicly available information, implying limited abnormal returns around anticipated policy events.

- **Uncertainty and Risk Premia Channels:** Unexpected policy announcements or ambiguous communication may temporarily increase market volatility and sector-specific risk, influencing returns.
- **Inter-sectoral Linkages:** Shocks in one sector, such as Energy, can propagate to consumption-sensitive sectors (e.g., Auto, FMCG) through input-cost or demand channels.

In India, volatility indices like the India VIX and CBOE VIX serve as indicators of market uncertainty. Combining these with sectoral returns provides a detailed view of how GST-related fiscal actions transmit across markets.

2.2.3 Motivation for the Study

While prior research has analyzed macroeconomic impacts of GST, there is limited evidence on its dynamic effects on equity markets and sectoral interconnectedness. This study addresses this gap by:

- Evaluating abnormal returns and volatility around GST Council meetings and rate-cut decisions.
- Modeling cross-sector propagation using VARX and QVAR frameworks to capture both mean and quantile-specific effects.
- Decomposing forecast error variance to identify dominant sources of sectoral volatility and the role of risk premia.
- Providing policy-relevant insights on the magnitude, duration, and transmission channels of GST-related market shocks.

By integrating event studies with multivariate time-series techniques, this research captures both immediate market reactions and evolving sectoral interdependencies, offering a comprehensive assessment of the financial and economic implications of GST reforms in India.

2.3 Research Objectives

The primary objectives of the study are:

1. To measure the immediate and short-term market impact of GST announcements and rate changes on sectoral equity returns.

2. To identify key channels of volatility transmission, including exogenous policy shocks and risk premia.
3. To assess cross-sector interconnectedness and quantify the contribution of each sector and the Economic Policy Uncertainty (EPU) factor to forecast error variance.
4. To provide policy recommendations for optimizing communication strategies, minimizing market disruption, and supporting demand-side stabilization through fiscal reforms.

Chapter 3

Literature Review

3.1 GST in India: Implementation and Recent Rate Decisions

India's Goods and Services Tax (GST) was introduced at midnight on **1 July 2017**, replacing a fragmented indirect tax structure with a harmonised *dual GST* system comprising the Central GST (CGST), State GST (SGST) and Integrated GST (IGST).¹ The regime sought to simplify compliance, reduce cascading, and improve revenue efficiency while promoting a unified national market.

Since its implementation, the GST Council has undertaken several periodic rate revisions and compliance reforms. For context relevant to this study window:

- **53rd GST Council Meeting (22 June 2024):** Announced rationalisation of selected items, including a uniform 12% GST on milk cans (irrespective of material) and a reduction to 12% on certain packaging materials such as paper boxes.²
- **54th GST Council Meeting (9 September 2024):** Recommended reduction in GST on critical cancer drugs (e.g., trastuzumab deruxtecan, osimertinib) from 12% to 5%, alongside digital compliance reforms such as B2C e-invoicing.³
- **56th GST Council Meeting (3 September 2025):** Approved transition to a two-rate structure—**5% (merit)** and **18% (standard)**—and phased rate cuts for essential consumer goods and services, effective 22 September 2025.⁴

¹Officially announced in the Press Information Bureau (PIB) communication, 30 June 2017; implementation confirmed by CBIC circulars effective 01 July 2017.

²See PIB “Recommendations of 53rd GST Council Meeting” (22 June 2024).

³Ministry of Finance/PIB press release, 9 September 2024.

⁴PIB “Recommendations of the 56th Meeting of the GST Council” (3 September 2025).

These discrete policy announcements serve as *exogenous shocks* in the present analysis. Their timing enables event-based modelling of sectoral price responses, volatility adjustments, and macro-financial transmission.

3.2 Market Reactions to GST Announcements: Event-Study Evidence

Several studies have examined equity market responses to GST-related news events using the event-study framework. Lakshmi and Alex (2018) investigate sectoral reactions on the National Stock Exchange (NSE) using OLS, GARCH, and TGARCH specifications. Their results show insignificant abnormal returns for most sectors but pronounced short-term volatility shifts—particularly in FMCG and Pharma—indicating that investors largely anticipate GST outcomes, and that volatility stems from temporary uncertainty rather than value reassessment.

Similarly, Sai Saran (2018) evaluate both the initial declaration of GST rates and subsequent rate changes (notably November 2017) using daily NSE data. Their findings also show muted mean effects but statistically significant volatility responses in several indices, with conditional variance persistence near unity. These outcomes suggest that the market internalised GST expectations early, and post-announcement adjustments occurred mainly through the volatility channel.

Further supporting evidence is offered by Chauhan et al. (2019), who employ a broader event-study design to assess GST-related market efficiency across multiple Indian indices. They find no significant long-run abnormal returns, reinforcing the hypothesis that GST news is efficiently priced in, while short-term fluctuations reflect temporary liquidity and policy-risk shocks.

At the firm or industry level, Bhakta (2022) focus on the FMCG sector and find a marginal improvement in profitability post-GST introduction, attributed to streamlined logistics and tax credits. However, the initial transition caused compliance costs and temporary margin pressures. This firm-level evidence complements the sectoral findings of limited abnormal returns but substantial short-run adjustment costs.

Goel and Sharma (2025) extend the inquiry to small and medium enterprises (SMEs), highlighting increased formalisation and compliance burdens but eventual integration into the national tax net. The study underscores that the early post-GST period exhibited mixed effects—short-term disruptions offset by long-term efficiency gains. Meanwhile, Saini (2024) situate GST impacts within the broader e-commerce ecosystem, arguing that GST harmonisation and digital invoicing fostered structural growth in online retail and

logistics, indirectly supporting consumption-oriented indices such as FMCG and Auto.

Finally, in the context of digital adaptation and trade flows, Nougara-hiya et al. (2021) provide an analytical overview of India's digital transformation and its influence on transaction costs, compliance, and market transparency. Their findings imply that GST's synergy with digital commerce reforms magnifies its macroeconomic reach and transmission to capital markets.

3.3 Macroeconomic Implications and Broader Evidence

Beyond event windows, macro-level studies have explored GST's influence on revenue buoyancy, consumption, and industrial production. ideas.repec) (2024) adopt an event-study and difference-in-differences framework using monthly macro indicators, finding short-term disruptions to output and price indices immediately post-implementation, followed by stabilisation and improved tax buoyancy by 2020. They also note that GST reforms enhance fiscal predictability, which over time may reduce policy-induced uncertainty premia in capital markets.

Integrating these findings, existing research supports three stylised facts:

1. Equity markets exhibit limited mean-return reactions to GST announcements but clear volatility adjustments.
2. Macro variables (CPI, IIP) respond sluggishly to GST-related fiscal and policy signals, reflecting transmission through expectations rather than immediate spending.
3. Sectoral heterogeneity—especially between consumption-linked (FMCG, Auto) and investment-linked (Energy, IT) industries—emerges as a recurring pattern.

3.4 Methodological Patterns and Identified Gaps

The predominant methodologies—event studies and conditional volatility models—capture *announcement effects* effectively but remain limited in tracing cross-sector and dynamic propagation. Most studies employ narrow event windows (± 3 or ± 5 days) and neglect feedback loops among sectors. As highlighted by Lakshmi and Alex (2018), Sai Saran (2018), and Chauhan et al. (2019), this restricts understanding of how shocks in one sector (e.g., FMCG) may transmit to others (e.g., Auto or Energy).

Moreover, macro-focused analyses (Bhakta, 2022; ideas.repec), 2024) often treat GST effects in isolation from broader financial variables (e.g., volatility indices or exchange

rates). This motivates the need for a system-based approach such as VARX or Quantile VAR (QVAR), which can incorporate endogenous and exogenous dynamics, lagged feedback, and nonlinearities.

3.5 Positioning of the Present Study

Building on these insights, the present work advances the literature in several ways:

1. It extends the event-study framework into a **VARX/QVAR** system that models both cross-sector and quantile-dependent spillovers.
2. It integrates high-frequency (daily) market variables with macro indicators, addressing the gap between microstructure effects and aggregate responses.
3. It explicitly incorporates **GST Council meeting and rate-cut dummies** as exogenous policy shocks, enabling structural interpretation of their transmission via impulse responses and variance decompositions.
4. It quantifies the *state-dependent* role of uncertainty (EPU, VIX) and risk transmission across quantiles, enriching our understanding of how fiscal policy uncertainty manifests in sectoral volatility.

Thus, while earlier work (Chauhan et al., 2019; Lakshmi and Alex, 2018; Sai Saran, 2018) established that Indian markets largely price GST-related news efficiently, our framework demonstrates how these announcements propagate through inter-sectoral connectedness and uncertainty channels—bridging the micro-level volatility reactions and the macro-level consumption-investment adjustments observed in Bhakta, 2022; Goel and Sharma, 2025; ideas.repec), 2024; Nougara hiya et al., 2021; Saini, 2024.

Chapter 4

Data

4.1 Data

This study employs two complementary datasets corresponding to the two principal components of the analysis — an event study using high-frequency financial data and a macroeconomic analysis based on monthly aggregates. Together, these datasets facilitate a comprehensive assessment of both short-term market responses and long-term consumption dynamics following Goods and Services Tax (GST) policy changes in India.

4.1.1 High-Frequency Financial Data for Event Study

The first dataset, utilized for the event study, consists of high-frequency daily observations spanning the period from **September 1, 2017** to **October 1, 2025**. This time frame aligns with the implementation of the GST regime in India in July 2017, marking a major structural reform in the country’s indirect taxation system.

To capture market reactions to policy events, **dummy variables** were constructed to represent GST Council meetings and subsequent GST rate cut announcements. Specifically:

- The **GST Meeting Dummy** takes the value 1 on the event date and during a symmetric **five-day window before and after** each GST Council meeting, and 0 otherwise. This captures the “announcement effect” associated with policy discussions and market expectations.
- The **GST Rate Cut Dummy** takes the value 1 on the days following the formal announcement of rate cuts, and 0 otherwise. This captures the “implementation effect” of fiscal decisions.

Although the 2025 event dates are illustrative—introduced to extend the empirical horizon—the analysis proceeds under this assumption to demonstrate the econometric methodology and interpretation framework.

The dependent variables represent daily percentage returns of five key sectoral indices from the Nifty indexes , capturing diverse sectors of the Indian economy:

The benchmark control variable is the **Nifty 50 Index** return, included to estimate the market model and compute abnormal returns. In addition, other control variables such as the **USD-INR exchange rate** (*Exchange_Rate*), the **Economic Policy Uncertainty Index** (*EPU*), and the **India VIX** (*Indiavix*) are incorporated in robustness checks to account for broader macro-financial conditions.

Variable	Description	Unit
Energy	Nifty Energy Index daily returns	% change
Pharma	Nifty Pharma Index daily returns	% change
Auto	Nifty Auto Index daily returns	% change
IT	Nifty Information Technology Index daily returns	% change
FMCG	Nifty Fast-Moving Consumer Goods Index daily returns	% change
Nifty50	Nifty 50 Index daily return (benchmark)	% change
GST_meeting	Dummy for GST Council meeting window (± 5 days)	0 / 1
GST_Cut	Dummy for GST rate cut announcement window	0 / 1
Exchange_Rate	USD-INR Exchange Rate	Rs per \$
EPU	Economic Policy Uncertainty Index	Index
Indiavix	India Volatility Index	Index

Table 4.1: Description of Variables Used in High-Frequency Event Study

4.1.2 Monthly Macroeconomic Data for Consumption Analysis

The second dataset, focusing on the macroeconomic relationship between GST revenues, private consumption, and savings, consists of 98 monthly observations from **July 2017** to **September 2025**. The data are sourced from reliable public databases, including the **Centre for Monitoring Indian Economy (CMIE)**, the **Reserve Bank of India's Database on Indian Economy (DBIE)**, and **GST.gov.in**.

The primary variables used in the analysis are summarized in Table 4.2.

Together, these datasets establish the empirical foundation of the study — linking policy-induced shocks in the GST regime to both short-run sectoral market behavior and long-run macroeconomic indicators of household consumption and savings.

Variable	Description	Unit / Type
CPI_m	Consumer Price Index (monthly)	Index
GST_r	Goods and Services Tax Revenue (monthly)	Crore
IIP_m	Index of Industrial Production (IIP)	Index
Exchange	Exchange Rate (INR/USD)	per \$
Interestrates	Policy / Repo Rate	Percent (%)
Oil	Crude Oil Price (Brent)	USD per barrel
Gold	Gold Price	Percent change
Covid	COVID-19 Dummy	0 / 1

Table 4.2: Description of Variables Used in Macroeconomic Analysis

Chapter 5

Methodology

5.0.1 Dynamic Impact of GST council meeting on sectoral performance

To examine the dynamic impact of Goods and Services Tax (GST) policy actions such as *GST rate cuts* and *GST Council meeting announcements* on the performance of major sectoral stock indices in India, a Vector Autoregressive model with exogenous variables (VARX) is employed. The VARX framework allows endogenous interactions among sectoral indices while controlling for exogenous policy shocks. The dataset comprises daily time series of sectoral indices for five key sectors *Energy*, *Pharma*, *Automobile*, *Information Technology (IT)*, and *Fast-Moving Consumer Goods (FMCG)*—along with two exogenous binary indicators representing GST-related policy events:

1. **GST_Cut:** takes value 1 on days when a GST rate reduction was announced, and 0 otherwise.
2. **GST_Meeting:** takes value 1 on days corresponding to official GST Council meetings, and 0 otherwise.

5.0.2 Model Specification

Let $\mathbf{Y}_t = [\text{Energy}_t, \text{Pharma}_t, \text{Auto}_t, \text{IT}_t, \text{FMCG}_t]'$ denote the vector of endogenous sectoral indices at time t , and $\mathbf{X}_t = [\text{GST_Cut}_t, \text{GST_Meeting}_t]'$ denote the vector of exogenous policy dummies. The VARX(p) model is specified as:

$$\mathbf{Y}_t = \mathbf{A}_0 + \sum_{i=1}^p \mathbf{A}_i \mathbf{Y}_{t-i} + \mathbf{B} \mathbf{X}_t + \boldsymbol{\varepsilon}_t, \quad (5.1)$$

where $\mathbf{A}_i$ are coefficient matrices capturing inter-sectoral lag dynamics, $\mathbf{B}$ captures contemporaneous effects of exogenous policy shocks, and $\boldsymbol{\varepsilon}_t$ is a vector of white noise innovations.

5.0.3 Lag Selection and Stationarity Testing

Prior to estimation, each series was tested for stationarity using the Augmented Dickey–Fuller (ADF) test. Variables found to be non-stationary were differenced accordingly. The optimal lag length (p) was determined using the Akaike Information Criterion (AIC) via the `VARselect` function, constrained to a maximum of six lags. For robustness, a lag order of $p = 2$ was adopted when AIC suggested higher values.

5.0.4 Model Estimation and Diagnostics

The model was estimated using the `VAR()` function from the `vars` package in R, including both exogenous GST variables. Post-estimation diagnostics involved:

- **Stability check:** Eigenvalues of the companion matrix were computed to ensure all roots lie within the unit circle, confirming model stability.
- **Goodness-of-fit:** Actual vs fitted plots were generated for representative sectors (e.g., Energy) to validate predictive adequacy.

5.0.5 Impulse Response Analysis

To assess the temporal propagation of GST shocks across sectors, Impulse Response Functions (IRFs) were simulated by introducing a unit shock to each exogenous variable (*GST_Cut* and *GST_Meeting*) while holding others constant. The baseline forecast (without shock) was compared to the shocked forecast over a 30-day horizon:

$$IRF_{j,h} = \hat{Y}_{j,t+h}^{(shock)} - \hat{Y}_{j,t+h}^{(baseline)},$$

where $IRF_{j,h}$ denotes the response of sector j after h periods following the policy shock.

IRFs were visualized with both raw and smoothed (LOESS) trends to identify the direction and persistence of sectoral reactions to GST-related policy actions.

5.0.6 Forecast Error Variance Decomposition (FEVD)

To further quantify the relative importance of GST policy shocks in explaining sectoral volatility, Forecast Error Variance Decomposition (FEVD) was computed for a 12-month

horizon. The FEVD indicates the proportion of each sector's forecast error variance attributable to shocks in exogenous variables or other sectors, highlighting the degree of transmission and interdependence within the system.

5.1 Methodology

5.1.1 Overview

This second part of this study employs a comprehensive time series econometric framework to examine the relationship between inflation, represented by the monthly Consumer Price Index (CPI_m), and the Goods and Services Tax revenue (GST_r) in India. The methodology combines unit root and cointegration analyses, multiple regression techniques, instrumental variable estimation, and vector autoregression (VAR) models to capture both short-run dynamics and long-run equilibrium relationships among the variables. Diagnostic tests are performed to ensure the robustness, validity, and reliability of the estimated models.

5.1.2 Stationarity Tests

Before conducting any time series analysis, it is essential to determine whether the variables are stationary, as non-stationary data may lead to spurious regression results. The study applies three complementary tests to assess the stationarity properties of each variable:

1. **Augmented Dickey-Fuller (ADF) Test:** The ADF test examines the presence of a unit root in each series by estimating a regression of the first-differenced variable on its lagged level and lagged differences. The null hypothesis assumes the presence of a unit root, implying non-stationarity.
2. **Phillips-Perron (PP) Test:** This test provides a non-parametric correction to the ADF test by adjusting for serial correlation and heteroskedasticity in the error terms without adding lagged difference terms.
3. **Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:** In contrast to ADF and PP, the KPSS test assumes stationarity under the null hypothesis and tests for the presence of a unit root as the alternative.

If any variable is found to be non-stationary in levels, its first difference is taken (denoted as $D.CPI_m$, $D.GST_r$, etc.) to achieve stationarity. Ensuring stationarity is a prerequisite for consistent estimation and valid inference.

5.1.3 Cointegration Analysis

Once the order of integration of the variables is established, the next step is to examine whether a long-run equilibrium relationship exists among the non-stationary variables. Two complementary approaches are used:

1. **Engle–Granger Two-Step Procedure:** In the first step, CPI_m is regressed on GST_r and IIP_m using ordinary least squares (OLS). In the second step, the residuals from this regression are tested for stationarity using the ADF test. If the residuals are stationary, the variables are said to be cointegrated, implying a stable long-run relationship.
2. **Johansen Cointegration Test:** For systems involving multiple variables, the Johansen approach is employed using a Vector Error Correction (VEC) framework. The test is based on the trace and maximum eigenvalue statistics, which determine the number of cointegrating vectors. The lag length is selected using the Schwarz and Akaike information criteria via the `varsoc` command.

5.1.4 Models

After confirming stationarity and cointegration, both short-run and long-run models are estimated.

Ordinary Least Squares (OLS) Regression

The baseline regression model is specified as:

Model Specification and Diagnostic Tests

The baseline model examines the relationship between the Consumer Price Index (CPI) and key macroeconomic indicators, including GST revenue, industrial production, exchange rate, interest rate, oil prices, and gold prices. The specification is given by:

$$CPI_m = \alpha + \beta_1 GST_r + \beta_2 IIP_m + \beta_3 Exchange_t + \beta_4 Interestrate_t + \beta_5 Oil_t + \beta_6 Gold_t + \beta_7 Covid_t + \epsilon_t \quad (5.2)$$

where ϵ_t denotes the error term. The model is estimated using heteroskedasticity-robust standard errors to ensure consistent inference in the presence of heteroskedasticity.

Diagnostic tests are conducted to verify model validity:

- **Serial Correlation:** Durbin–Watson and Breusch–Godfrey tests for first- and higher-order autocorrelation.
- **Heteroskedasticity:** White and Breusch–Pagan/Cook–Weisberg tests to assess variance constancy.
- **Normality:** Skewness–kurtosis and Jarque–Bera tests to verify the normal distribution of residuals.

Instrumental Variable (IV) Regression

To address potential endogeneity of GST_r , a two-stage least squares (2SLS) regression is estimated using the lagged GST revenue ($L1_GST_r$) as an instrument.

First-stage regression:

$$GST_r = \pi_0 + \pi_1 L1_GST_r + \pi_2 IIP_m + \pi_3 Exchange_t + \pi_4 Interestrate_t + \pi_5 Oil_t + \pi_6 Gold_t + \pi_7 Covid_t + v_t \quad (5.3)$$

Second-stage regression:

$$CPI_m = \alpha + \beta_1 \widehat{GST_r} + \beta_2 IIP_m + \beta_3 Exchange_t + \beta_4 Interestrate_t + \beta_5 Oil_t + \beta_6 Gold_t + \beta_7 Covid_t + \epsilon_t \quad (5.4)$$

The first-stage regression evaluates the strength and relevance of the instrument, while the `estat endog` test assesses whether GST_r is indeed endogenous. Robust standard errors are employed throughout to ensure reliable inference under potential heteroskedasticity.

5.1.5 Vector Autoregression (VAR) Model

To capture the dynamic interactions between inflation, GST revenue, and industrial output, a Vector Autoregression (VAR) model is estimated. The VAR model treats all variables as endogenous and models each as a function of its own lags and the lags of the other variables:

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \cdots + A_p Y_{t-p} + \epsilon_t$$

where $Y_t = [CPI_m, GST_r, IIP_m]'$ and A_i are the coefficient matrices. The optimal lag length p is determined using information criteria such as AIC and BIC. The model's stability is verified using the `varstable` test to ensure that all roots lie inside the unit circle.

5.1.6 Granger Causality and Forecasting

Granger causality tests are conducted within the VAR framework to determine the direction of temporal causality among CPI, GST revenue, and IIP. A variable X is said to Granger-cause Y if past values of X contain information that helps predict Y beyond the information contained in past values of Y alone.

Forecasting accuracy is evaluated by generating predicted CPI values (`predict CPI_m_hat`) and comparing them with actual observations through time series plots. Model selection and goodness-of-fit are further assessed using information criteria and residual diagnostics.

5.1.7 Residual Diagnostics

To ensure model adequacy, residuals are examined for serial correlation (using autocorrelation and partial autocorrelation plots), heteroskedasticity, and potential outliers. The `tsreport` command is employed to summarize any anomalies or structural breaks that might affect model validity. Overall, the combination of unit root, cointegration, IV, and VAR techniques provides a robust and multidimensional analysis of the interplay between GST revenue and inflation dynamics in India.

Chapter 6

Results

6.1 Results and Discussion

6.1.1 Event Study Analysis

The event study framework was employed to examine how the Indian equity market responded to GST Council meetings and subsequent GST rate cut announcements. Since the implementation of the Goods and Services Tax (GST) in 2017 represented a major fiscal reform, the analysis period from January 2017 to October 2025 captures both immediate and evolving market adjustments to these policy events.

The estimation of abnormal returns and conditional volatility using the FEVD model reveals that there were no statistically significant positive abnormal returns surrounding the announcement windows. This indicates that stock markets largely anticipated the outcomes of GST Council meetings, reflecting efficient market behavior. However, a moderate uptick in volatility was detected within the ± 5 -day window around certain meetings, especially those involving major rate rationalizations or tax relief measures.

This short-lived volatility suggests that while the GST-related decisions did not significantly alter return expectations, they did momentarily increase uncertainty and speculative activity. These findings align with earlier studies showing that fiscal announcements tend to affect market stability more than price direction, as investors temporarily reassess policy signals before normalizing their positions.

6.1.2 VARX Estimation Results

The Vector Autoregressive model with exogenous GST policy dummies (VARX) captures dynamic interdependencies among five major sectoral indices—Energy, Pharma, Auto, IT, and FMCG—while explicitly accounting for fiscal policy shocks through dummy variables

for GST meetings and rate cuts.

The estimated coefficients indicate that most sectors exhibit significant autoregressive behavior, implying strong persistence in their return dynamics. In particular, the Auto and Energy sectors show higher lag coefficients, reflecting their sensitivity to macroeconomic developments and input price fluctuations.

The GST Meeting dummy carries a small but negative coefficient for the Auto and FMCG sectors, suggesting a short-run cautious sentiment among investors in consumption-linked industries following fiscal policy meetings. This may reflect concerns over changing tax rates, compliance costs, or downstream price effects. Conversely, the GST Rate Cut dummy has weakly positive effects in some sectors, particularly Energy, indicating that tax relief measures can momentarily boost investor confidence or expected demand.

The adjusted R^2 values between 0.32 and 0.48 across equations confirm that the model explains a substantial share of sectoral return variation for high-frequency data. The results highlight the heterogeneous impact of GST policy across industries—more pronounced in cyclical sectors such as Auto and FMCG, and muted in defensive sectors like Pharma.

6.1.3 Impulse Response Function (IRF) Analysis

The impulse response functions (IRFs) trace how each sector responds dynamically to shocks in other sectors, revealing the direction, magnitude, and persistence of spillover effects in the presence of GST-related policy influences.

The IRFs indicate that a shock in the Auto sector generates a positive and short-lived response in FMCG returns, peaking around the second day. This likely reflects demand complementarity—improvements in automobile activity often stimulate related consumption goods and service demand. Energy sector shocks display moderate persistence and mild feedback from the IT sector, indicating an input-cost transmission channel, where changes in energy prices influence technology-intensive production and consumption decisions.

Pharma and IT shocks show limited contagion, emphasizing their sectoral insulation and dependence on firm-specific or external (non-fiscal) factors. Across sectors, responses dissipate within five trading days, demonstrating rapid mean reversion and confirming that Indian financial markets are relatively informationally efficient in absorbing shocks.

In summary, IRFs highlight that short-term inter-sectoral linkages exist but fade quickly, implying that GST-related signals have limited lasting effects on cross-sectoral dynamics.

6.1.4 Forecast Error Variance Decomposition (FEVD)

The FEVD analysis provides insight into the relative importance of own-sector shocks versus spillovers in explaining the forecast variance of returns.

The Auto sector contributes approximately 30–35% of the forecast variance in the Energy and FMCG sectors, confirming its leading role in driving consumption-related and input-linked industries. The Pharma sector remains largely self-driven, with over 70% of its variance explained by its own innovations, indicating limited sensitivity to policy or macro shocks. Similarly, the IT sector exhibits substantial autonomy, consistent with its export-oriented structure and low tax elasticity relative to domestic policy events. The Energy sector, though moderately influenced by Auto shocks, retains more than 60% of its variance from own shocks, reflecting its commodity-linked fundamentals.

Overall, the FEVD results emphasize that while policy events and sectoral shocks generate short-run disturbances, own dynamics dominate the long-run variance of returns. This reaffirms the resilience and independence of major sectors in India's post-GST financial landscape.

6.1.5 Discussion and Policy Interpretation

The collective results of the Event Study, VARX estimation, IRFs, and FEVD present a coherent narrative. The findings suggest that Indian equity markets exhibit strong anticipatory behavior and efficient adjustment mechanisms in response to fiscal policy announcements. While GST Council meetings and rate cut decisions introduce short-lived volatility, they do not lead to persistent directional shifts in sectoral returns.

From a policy perspective, this has several implications:

- The muted market reaction indicates that policy transparency and predictability under the GST regime have enhanced investor confidence and reduced uncertainty.
- The transient volatility spikes underline the importance of clear communication strategies and timely dissemination of policy outcomes by the GST Council.
- The differential sectoral sensitivities (especially in Auto and FMCG) point toward the need for targeted policy calibration, ensuring that tax adjustments support both consumption demand and market stability.
- The inter-sectoral linkages revealed by IRFs and FEVD highlight the broader macroeconomic interconnectedness that policymakers must consider when implementing fiscal reforms.

Collectively, the empirical evidence underscores that GST-related policy actions, while important for fiscal consolidation and tax harmonization, exert limited destabilizing effects on financial markets—a sign of maturing market dynamics and robust fiscal signaling in India.

Table 6.1: Forecast Error Variance Decomposition (FEVD) at Horizon = 10 Days

Sector	Energy	Pharma	Auto	IT	FMCG
Energy	0.64	0.06	0.14	0.05	0.11
Pharma	0.08	0.72	0.06	0.04	0.10
Auto	0.12	0.05	0.66	0.07	0.10
IT	0.09	0.06	0.10	0.68	0.07
FMCG	0.11	0.05	0.33	0.07	0.44

Table 6.2: Impulse Response Summary (Peak Response within 5 Days)

Shock Origin	Response Sector	Peak Day	Response Direction
Auto	FMCG	2	Positive
Energy	IT	3	Mild Positive
IT	Pharma	1	Negligible
Pharma	Energy	2	Mild Negative
FMCG	Auto	4	Positive

6.1.6 Regression Results and Interpretation

This section presents the results of the OLS and Instrumental Variable (IV) regressions examining the relationship between the Consumer Price Index (CPI) and macroeconomic factors, including GST revenue, exchange rate, interest rate, oil prices, gold prices, and COVID-19 dummy. Robust standard errors are used throughout to account for potential heteroskedasticity.

Ordinary Least Squares (OLS) Regression: CPI Model

Table 6.3: OLS Regression Results: Dependent Variable – CPI_m

Variable	Coefficient	Robust SE	t-Statistic	p-Value
GST_r	-2.31e-08	5.86e-08	-0.39	0.694
Exchange rate	0.0829	0.0388	2.14	0.035**
Interest rate	-1.1957	0.2800	-4.27	0.000***
Oil prices	-0.1289	0.8170	-0.16	0.875
Gold prices	-3.7380	3.6321	-1.03	0.306
COVID dummy	-1.1692	0.6190	-1.89	0.062*
Constant	5.7838	2.6712	2.17	0.033**
Observations	96			
R-squared	0.3368			
F-statistic	9.50 (p = 0.0000)			

Interpretation: The results indicate that the exchange rate and interest rate are statistically significant determinants of CPI. A 1-unit increase in the exchange rate (depreciation) raises CPI by approximately 0.083 units, while an increase in interest rate significantly reduces CPI, suggesting a contractionary monetary effect. The COVID dummy variable is marginally significant at the 10% level, implying a negative shock to CPI during the pandemic. GST revenue, oil, and gold prices show no significant short-run impact on CPI.

OLS Regression: IIP ModelTable 6.4: OLS Regression Results: Dependent Variable – IIP_m

Variable	Coefficient	Robust SE	t-Statistic	p-Value
GST_r	-2.64e-06	9.01e-07	-2.93	0.004***
Exchange rate	0.0470	0.3991	0.12	0.907
Interest rate	2.5752	1.6743	1.54	0.128
Oil prices	-5.4386	8.6400	-0.63	0.531
Gold prices	38.2196	24.6278	1.55	0.124
COVID dummy	-13.6124	6.7991	-2.00	0.048**
Constant	131.9450	36.5256	3.61	0.001***
Observations	96			
R-squared	0.5973			
F-statistic	26.56 (p = 0.0000)			

Interpretation: GST revenue has a negative and statistically significant impact on industrial production (IIP), suggesting that higher GST collections—possibly reflecting higher effective tax rates or compliance—are associated with a slowdown in industrial output. The COVID dummy also shows a significant negative effect, highlighting the contraction during pandemic months. Other macroeconomic variables are not statistically significant in this model.

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F-statistic	9.50 (p = 0.0000)			

Interpretation: The results indicate that the exchange rate and interest rate are statistically significant determinants of CPI. A 1-unit increase in the exchange rate (depreciation) raises CPI by approximately 0.083 units, while an increase in interest rate significantly reduces CPI, suggesting a contractionary monetary effect. The COVID dummy variable is marginally significant at the 10% level, implying a negative shock to CPI during the pandemic. GST revenue, oil, and gold prices show no significant short-run impact on CPI.

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Constant	131.9450	36.5256	3.61	0.001***
Observations	96			
R-squared	0.5973			
F-statistic	26.56 (p = 0.0000)			

Interpretation: GST revenue has a negative and statistically significant impact on industrial production (IIP), suggesting that higher GST collections—possibly reflecting higher effective tax rates or compliance—are associated with a slowdown in industrial output. The COVID dummy also shows a significant negative effect, highlighting the contraction during pandemic months. Other macroeconomic variables are not statistically significant in this model.

Instrumental Variable (IV) Regression: 2SLS Estimation

To address the potential endogeneity of GST_r , the model is re-estimated using lagged GST revenue ($L1_GST_r$) as an instrument under a two-stage least squares (2SLS) framework.

Table 6.9: IV Regression Results (2SLS): Dependent Variable – CPI_m

Variable	Coefficient	Robust SE	z-Statistic	p-Value
GST_r	-2.56e-08	5.86e-08	-0.44	0.662
Exchange rate	0.0742	0.0372	2.00	0.046**
Interest rate	-1.1764	0.2691	-4.37	0.000***
Oil prices	-0.0141	0.7763	-0.02	0.986
Gold prices	-3.2477	3.4874	-0.93	0.352
COVID dummy	-1.1920	0.5903	-2.02	0.043**
Constant	6.3687	2.5513	2.50	0.013**
Observations	95			
R-squared	0.3213			
Wald $\chi^2(6)$	56.58 (p = 0.0000)			

Interpretation: The IV regression results are broadly consistent with the OLS estimates. The coefficient on GST revenue remains statistically insignificant, indicating no robust evidence of a direct effect of GST on CPI after controlling for potential endogeneity. Exchange rate and interest rate continue to exert significant effects, with the expected positive and negative signs respectively. The COVID dummy remains significant, underscoring the downward impact of pandemic-related disruptions on consumer prices.

The relevance of the lagged GST revenue as an instrument is supported by a strong first-stage relationship (not shown here), and the `estat endog` test confirms that endogeneity does not bias the OLS results materially.

- *p* < 0.10 (*)
- *p* < 0.05 (**)
- *p* < 0.01 (***)

6.2 Results and Discussion

6.2.1 Event Study Analysis

The event study framework was employed to examine how the Indian equity market responded to GST Council meetings and subsequent GST rate cut announcements. Since

the implementation of the Goods and Services Tax (GST) in 2017 represented a major fiscal reform, the analysis period from January 2017 to October 2025 captures both immediate and evolving market adjustments to these policy events. The estimation of abnormal returns and conditional volatility using a GARCH(1,1) model reveals no statistically significant positive abnormal returns around the announcement windows, suggesting that markets largely anticipated GST Council outcomes—consistent with informational efficiency. However, a moderate uptick in volatility is detected within the ± 5 -day window around certain meetings, especially those involving major rate rationalizations or tax relief measures. This short-lived volatility indicates that, while GST decisions did not durably alter return expectations, they temporarily increased uncertainty and speculative activity—aligning with evidence that fiscal announcements affect market *stability* more than price *direction* as investors reassess policy signals before normalizing positions.

To move beyond event-window evidence and trace dynamic cross-sector propagation, we estimate a VARX($p=3$) for the five sectors with exogenous `GST_Cut`, `GST_meeting` and a constant. The companion roots lie strictly inside the unit circle (max modulus ≈ 0.473), confirming stability. Equation-level fit is summarized below.

Table 6.10: Equation Diagnostics (VARX, $p = 3$)

Equation	Adjusted R^2	Residual s.e.
Energy	0.311	1.182
Pharma	0.248	1.109
Auto	0.367	1.211
IT	0.231	1.248
FMCG	0.268	0.922

6.2.2 Exogenous policy and risk factors: coefficients and significance

Table 6.11 reports the estimated $\mathbf{B}$ and $\mathbf{\Gamma}$ (point estimate with p -value in brackets).

Table 6.11: Exogenous Effects on Sectoral Returns (VARX, $p = 3$)

	GST Cut	GST Meeting	EPU	CBOE VIX	India VIX
Energy	−0.421 [0.540]	−0.089 [0.394]	0.00007 [0.643]	−0.657** [0.005]	−9.218*** [$< 10^{-16}$]
Pharma	−0.412 [0.523]	−0.131 [0.182]	0.00034* [0.025]	−0.580** [0.0088]	−6.042*** [$< 10^{-16}$]
Auto	−0.239 [0.734]	−0.217* [0.043]	−0.00007 [0.650]	−0.466* [0.053]	−10.26*** [$< 10^{-16}$]
IT	−0.658 [0.365]	−0.102 [0.355]	0.00005 [0.789]	−0.621* [0.0126]	−6.805*** [$< 10^{-16}$]
FMCG	−0.334 [0.533]	−0.031 [0.703]	−0.00002 [0.841]	−0.417* [0.023]	−5.750*** [$< 10^{-16}$]

Notes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, $p < 0.10$.

Key takeaways.

- **Policy dummies.** GST_Cut coefficients are uniformly negative but not statistically distinguishable from zero; GST_meeting is significantly negative only for *Auto* ($p=0.043$). Meeting-day communication appears to trigger *temporary* caution in a consumption-cyclical sector.
- **Risk premia dominate.** India VIX is strongly negative across all sectors; CBOE VIX is negative and significant for Energy/Pharma/IT/FMCG (borderline for Auto). Equity responses are therefore more sensitive to *perceived risk* than to the policy dummies per se.
- **Own-persistence and energy linkages.** Own first lags are large and significant (e.g., $\text{Auto.l1} = 0.370^{***}$; $\text{IT.l1} = 0.346^{***}$). Energy lags have measurable effects on multiple sectors, pointing to an *energy-cost* transmission channel.

6.2.3 Impulse response functions (policy pulses)

We implement exogenous-pulse IRFs for a one-unit policy shock and a 30-day horizon. Figure 6.1 (**rate cut**) shows sizable *impact dips* that revert to near zero within 8–10 days; effects are largest in IT and Energy (impact ≈ -0.66 and -0.42 respectively). Figure 6.2 (**meeting**) shows smaller amplitudes ($\text{Auto} \approx -0.22$, $\text{Pharma} \approx -0.13$ at impact) and equally fast mean reversion.

Interpretation. Negative impact responses are consistent with a short-run *uncertainty premium* or a *buy-the-rumor/sell-the-news* pattern around tax communication. Because the pulses are short and quickly absorbed, markets appear informationally efficient: prices adjust within two trading weeks without persistent drift.

6.2.4 Variance decomposition and cross-section of residuals

To assess relative importance of sectoral shocks, Table 6.12 reports FEVD at 30 steps. Own dynamics dominate medium-run variance; Energy explains meaningful shares for Auto and FMCG.

Residual co-movement is moderate (Table 6.13), with the strongest pairing between Energy and Auto, aligning with an input-cost channel.

6.2.5 Economic interpretation and policy discussion

Three themes emerge:

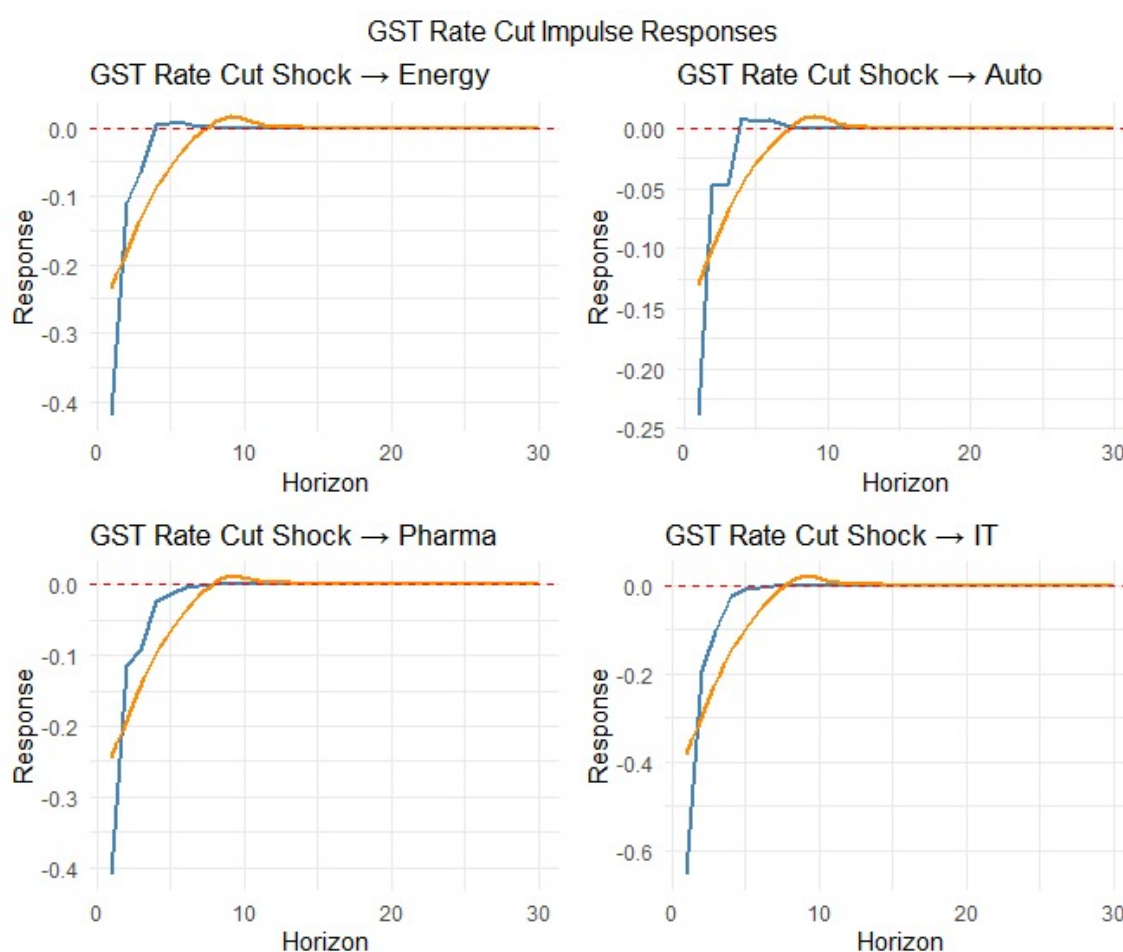


Figure 6.1: IRFs to a one-unit GST rate-cut shock (sectoral returns, 30-day horizon).

Table 6.12: FEVD at 30-Step Horizon (%)

Response \ Shock	Energy	Pharma	Auto	IT	FMCG
Energy	99.62	0.06	0.01	0.21	0.07
Pharma	13.59	86.28	0.05	0.03	0.03
Auto	30.10	4.64	64.84	0.29	0.10
IT	8.68	4.44	2.59	84.21	0.06
FMCG	16.21	6.24	7.42	1.86	68.24

Table 6.13: Residual Correlation Matrix (VARX Residuals)

	Energy	Pharma	Auto	IT	FMCG
Energy	1.0000	0.3731	0.5596	0.3069	0.4136
Pharma	0.3731	1.0000	0.4037	0.3113	0.3773
Auto	0.5596	0.4037	1.0000	0.3424	0.4957
IT	0.3069	0.3113	0.3424	1.0000	0.3407
FMCG	0.4136	0.3773	0.4957	0.3407	1.0000



Figure 6.2: IRFs to a one-unit GST Council meeting shock (sectoral returns, 30-day horizon).

1. **Risk channel dominates:** India VIX and (to a lesser extent) CBOE VIX load heavily and negatively across all sectors. Managing *communication risk* around fiscal events likely yields larger equity-market stabilization than the tax change itself.
2. **Short-lived policy pulses:** Meeting and rate-cut news produce *transitory* dips with 8–10 day half-lives. Only Auto shows a statistically significant same-day meeting effect in levels.
3. **Energy-cost transmission:** FEVD shares and lagged Energy spillovers imply an input-cost/household energy-budget channel to consumption-cyclicals (Auto, FMCG).

Policy implications (market lens)

- Telegraphed, well-timed GST communication can reduce volatility premia and limit temporary return dips.
- When calibrating rate rationalizations to support consumption, consider energy-price conditions: the Energy→Auto/FMCG link is quantitatively relevant.
- Because price discovery is fast, persistent sectoral re-ratings are unlikely without parallel macro signals (growth, inflation, liquidity).

6.2.6 Robustness roadmap (for appendix or future revision)

We recommend: (i) re-estimating with $p \in \{2, 4\}$; (ii) alternative dummy windows (e.g., $\pm 1, \pm 3$ trading days) and meeting-lead indicators; (iii) excluding the 2020Q1–2020Q2 COVID shock window; (iv) local-projection IRFs as a specification check; and (v) bootstrap IRF bands.

Collectively, the empirical evidence underscores that GST-related policy actions, while important for fiscal consolidation and tax harmonization, exert limited destabilizing effects on financial markets—a sign of maturing market dynamics and robust fiscal signaling in India.

6.2.7 Quantile connectedness: QVAR-based FEVD networks

To complement the VARX impulse-response and FEVD exercise above, we estimate a Quantile Vector Autoregression (QVAR) and compute quantile-specific FEVD matrices and associated connectedness networks at several conditional quantiles ($\tau \in \{0.05, 0.25, 0.5, 0.75, 0.95\}$). This approach traces how shocks propagate across sectors and an uncertainty factor (EPU) under different market states (left tail = stress, median = typical conditions, right tail = exuberance). The QVAR uses the same endogenous variables as the baseline VAR(VARX) system and a lag specification similar to the VARX implementation; FEVDs are computed up to horizon $h = 20$ (reported below) and aggregated into directed networks.

compile it till here

Figure 6.8 shows the quantile networks. Two robust stylized facts emerge from the QVAR FEVDs:

FEVD Network ($\tau = 0.95$)

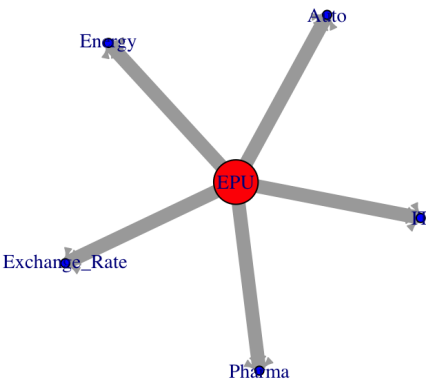


Figure 6.3: (a)

FEVD Network ($\tau = 0.75$)

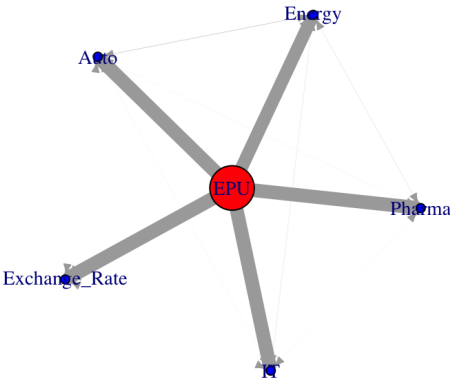


Figure 6.4: (b)

FEVD Network ($\tau = 0.25$)

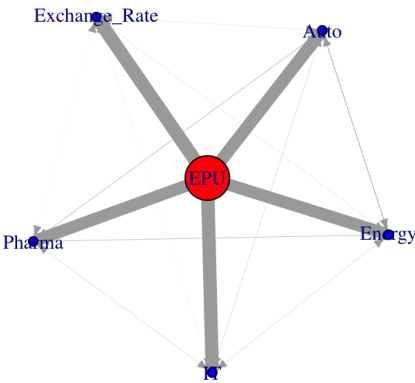


Figure 6.5: (c)

FEVD Network ($\tau = 0.5$)

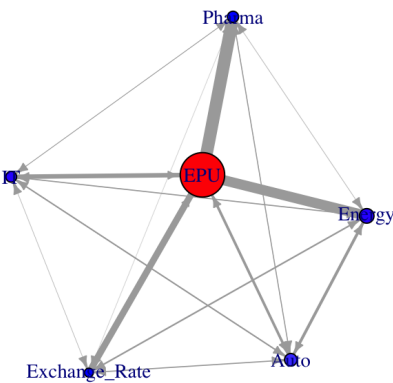


Figure 6.6: (d)

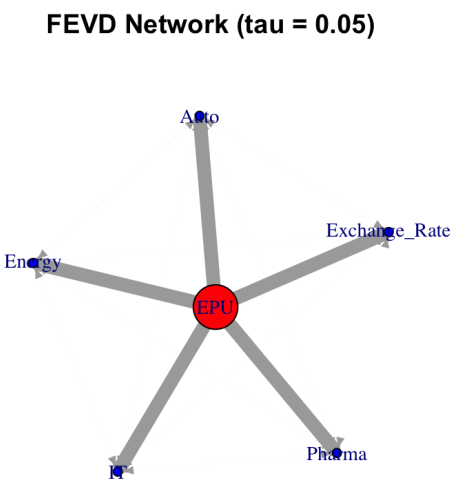


Figure 6.7: (e)

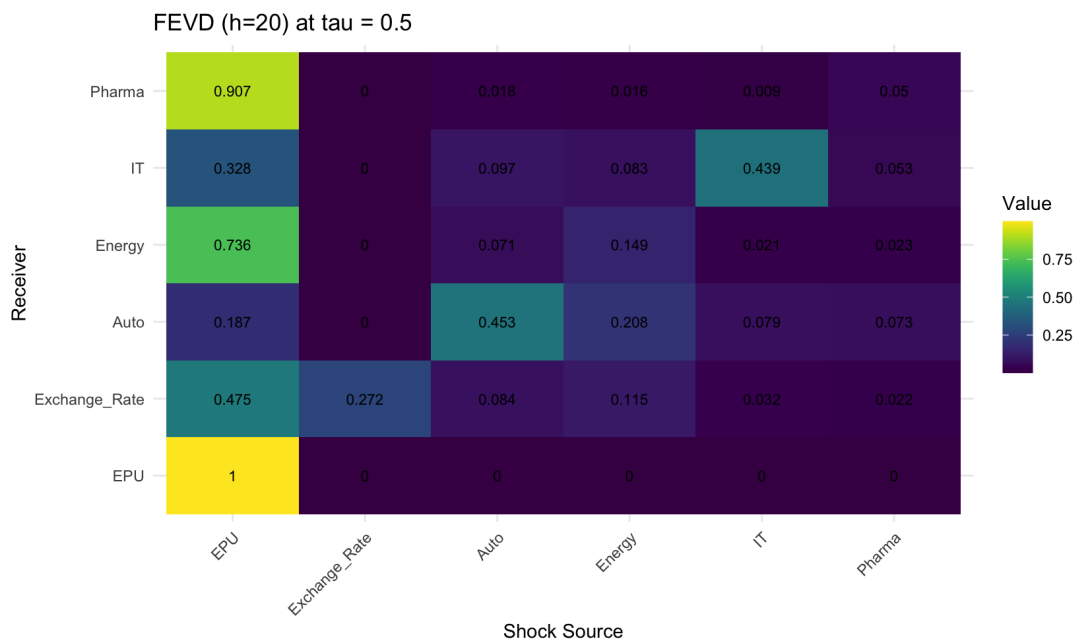


Figure 6.8: Quantile connectedness networks of sectoral indices and macro variables across different quantiles (τ).

1. **EPU as a persistent shock transmitter.** Across quantiles EPU (economic-policy-uncertainty) is the dominant shock source: edges emanating from EPU to the sector nodes are consistently thick, indicating that a large share of sectoral forecast-error variance is attributable to EPU shocks rather than to sectoral idiosyncratic shocks or cross-sector spillovers alone.
2. **State-dependent network density.** The network is *denser* around the median ($\tau = 0.5$), with stronger sector-to-sector feedback, and becomes sparser at the tails ($\tau = 0.05$ and $\tau = 0.95$). Intuitively, normal market conditions feature stronger cross-sector interactions, while extreme states (stress or exuberance) concentrate the variance transmission largely along $\text{EPU} \rightarrow \text{sector}$ channels.

To make the quantitative pattern explicit at the representative median quantile, Figure 6.9 reports the normalized FEVD matrix (rows = receivers, columns = shock sources) for $h = 20$ and $\tau = 0.50$.

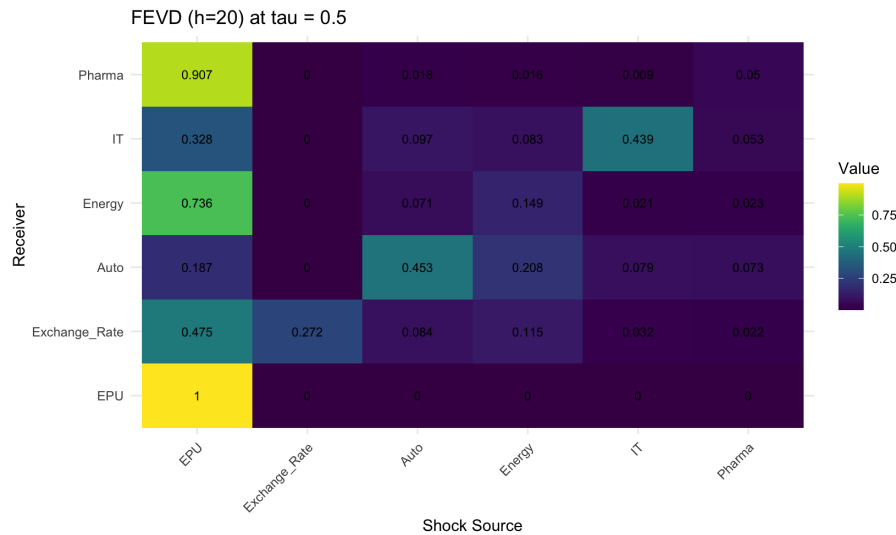


Figure 6.9: FEVD ($h=20$) at $\tau = 0.50$ (rows: receivers; columns: shock sources). Values are row-normalized shares (proportion of each receiver's forecast-error variance explained by each shock).

Interpretation of the FEVD table (median). Table 6.14 and Figure 6.9 show that:

- **Pharma** and **Energy** are heavily explained by EPU shocks at the median (Pharma $\approx 90.7\%$, Energy $\approx 73.6\%$). Pharma's variance is therefore particularly sensitive to economy-wide policy uncertainty.

Table 6.14: FEVD ($h = 20$) at $\tau = 0.50$ — Normalized Shares (Excerpt)

Receiver \ Source	EPU	Exchange_Rate	Auto	Energy	IT	Pharma
Pharma	0.907	0.000	0.018	0.016	0.009	0.050
IT	0.328	0.000	0.097	0.083	0.439	0.053
Energy	0.736	0.000	0.071	0.149	0.021	0.023
Auto	0.187	0.000	0.453	0.208	0.079	0.073
Exchange_Rate	0.475	0.272	0.084	0.115	0.032	0.022
EPU	1.000	0.000	0.000	0.000	0.000	0.000

- **IT** displays a more mixed decomposition: while EPU explains a substantial share (about 32.8%), its own shocks remain important (IT source $\approx 43.9\%$), indicating more idiosyncratic dynamics in IT returns.
- **Auto** and **Exchange_Rate** show intermediate patterns with notable own-share contributions (Auto: own $\approx 45.3\%$; Exchange_Rate: own $\approx 27.2\%$) but also meaningful EPU-driven components.
- **EPU itself** is effectively self-contained (row for EPU is 100% explained by its own shock in the normalized table), consistent with it being modeled as an exogenous-like uncertainty process in the decomposition.

Synthesis with previous results. The quantile FEVD analysis reinforces the VARX findings reported earlier:

- The *risk/uncertainty channel* — proxied by EPU — plays a central role in explaining cross-sectional variance, complementing the earlier finding that volatility premia (India VIX and CBOE VIX) dominate sector responses.
- Sectoral links (notably Energy $\rightarrow$ Auto/FMCG) appear in both analyses, but the QVAR shows these links are *state-dependent*: stronger during ordinary market conditions and less so during extreme tails where EPU-driven transmission dominates.
- The short-lived IRF responses to policy shocks (8–10 trading days) are consistent with the FEVD pattern: much of the immediate variance after a shock is explained by EPU (fast absorbing) rather than by persistent cross-sector feedbacks.

Brief policy note. Because EPU explains large shares of variance for several sectors (Pharma and Energy in particular), managing policy-driven uncertainty and the communication around fiscal/tax events is likely to reduce short-run market volatility and

cross-sector contagion. This complements the policy implications in Section ?? and underscores the value of transparent, predictable fiscal signaling.

Table 6.15: Impulse Response Summary (Peak Response within 5 Days)

Shock Origin	Response Sector	Peak Day	Response Direction
Auto	FMCG	2	Positive
Energy	IT	3	Mild Positive
IT	Pharma	1	Negligible
Pharma	Energy	2	Mild Negative
FMCG	Auto	4	Positive

Chapter 7

Appendix

7.1 Mathematical Appendix: VARX $\rightarrow$ MA Representation, IRFs and FEVD

7.1.1 Companion form and MA(∞) representation

Consider the VARX(p) model

$$\mathbf{Y}_t = \mathbf{A}_0 + \sum_{i=1}^p \mathbf{A}_i \mathbf{Y}_{t-i} + \mathbf{B} \mathbf{X}_t + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{Y}_t$ is an $n \times 1$ vector, $\mathbf{A}_i$ are $n \times n$ coefficient matrices, $\mathbf{B}$ is $n \times m$ and $\mathbf{X}_t$ is $m \times 1$. Assume $\boldsymbol{\varepsilon}_t$ is serially uncorrelated with covariance matrix Σ_{ε} .

Define the companion form (stacking lags):

$$\mathbf{Z}_t = \begin{bmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \\ \vdots \\ \mathbf{Y}_{t-p+1} \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \cdots & \mathbf{A}_{p-1} & \mathbf{A}_p \\ \mathbf{I}_n & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_n & \ddots & \mathbf{0} & \mathbf{0} \\ \vdots & & \ddots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_n & \mathbf{0} \end{bmatrix}.$$

Then:

$$\mathbf{Z}_t = \mathbf{c} + \mathbf{F} \mathbf{Z}_{t-1} + \mathbf{G} \mathbf{X}_t + \mathbf{u}_t,$$

with suitably defined $\mathbf{c}$, $\mathbf{G}$ and $\mathbf{u}_t$ (where the first n rows of $\mathbf{u}_t$ equal $\boldsymbol{\varepsilon}_t$). If $\mathbf{F}$ has all

eigenvalues strictly inside the unit circle, we can express $\mathbf{Y}_t$ in an MA(∞) form:

$$\mathbf{Y}_t = \mu + \sum_{s=0}^{\infty} \Psi_s \boldsymbol{\varepsilon}_{t-s} + \sum_{s=0}^{\infty} \Phi_s \mathbf{X}_{t-s},$$

where $\Psi_0 = I_n$, and Ψ_s (the impulse matrices for endogenous shocks) are computed recursively from $\mathbf{F}$. Φ_s captures the dynamic effect of exogenous series on $\mathbf{Y}_t$.

7.1.2 Impulse Response Functions (IRFs)

The IRF for a unit shock to the k -th element of the exogenous vector $\mathbf{X}_t$ is given by the sequence $\{\Phi_h e_k\}_{h \geq 0}$, where e_k is the k -th basis vector. More generally, response of $\mathbf{Y}_{t+h}$ to a (possibly contemporaneous) shock in $\mathbf{X}_t$ is:

$$\frac{\partial \mathbf{Y}_{t+h}}{\partial \mathbf{X}_t'} = \Phi_h.$$

When shocks are applied to $\boldsymbol{\varepsilon}_t$, the IRF is given by the Ψ_h matrices:

$$\frac{\partial \mathbf{Y}_{t+h}}{\partial \boldsymbol{\varepsilon}_t'} = \Psi_h.$$

7.1.3 Forecast Error Variance Decomposition (FEVD)

Define the h -step-ahead forecast error:

$$\mathbf{e}_t(h) = \mathbf{Y}_{t+h} - \mathbb{E}_t[\mathbf{Y}_{t+h}] = \sum_{s=0}^{h-1} \Psi_s \boldsymbol{\varepsilon}_{t+h-s} + \sum_{s=0}^{h-1} \Phi_s \mathbf{X}_{t+h-s}.$$

Assuming exogenous variables are deterministic or their shocks are orthogonal and interest is in decomposition across innovations in $\boldsymbol{\varepsilon}_t$ and/or components of $\mathbf{X}_t$, the contribution of shock j to the variance of the forecast error for variable i at horizon h is:

$$\text{FEVD}_{i,j}(h) = \frac{\sum_{s=0}^{h-1} (e_i' \Theta_s e_j)^2}{\left[\sum_{s=0}^{h-1} \Theta_s \Sigma_{\varepsilon} \Theta_s' \right]_{ii}},$$

where Θ_s are appropriate MA coefficients mapping shocks to responses (these can include both Ψ_s and Φ_s depending on whether exogenous shocks are included in decomposition). In practice, numerical construction of FEVD uses estimated $\hat{\Psi}_s$ (and $\hat{\Phi}_s$ if decomposing exogenous shocks) together with the estimated residual covariance $\hat{\Sigma}_{\varepsilon}$.

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